



IT479M Machine Learning Minor Project

Identification of Encryption Algorithms and Plaintext Recovery Using Machine Learning

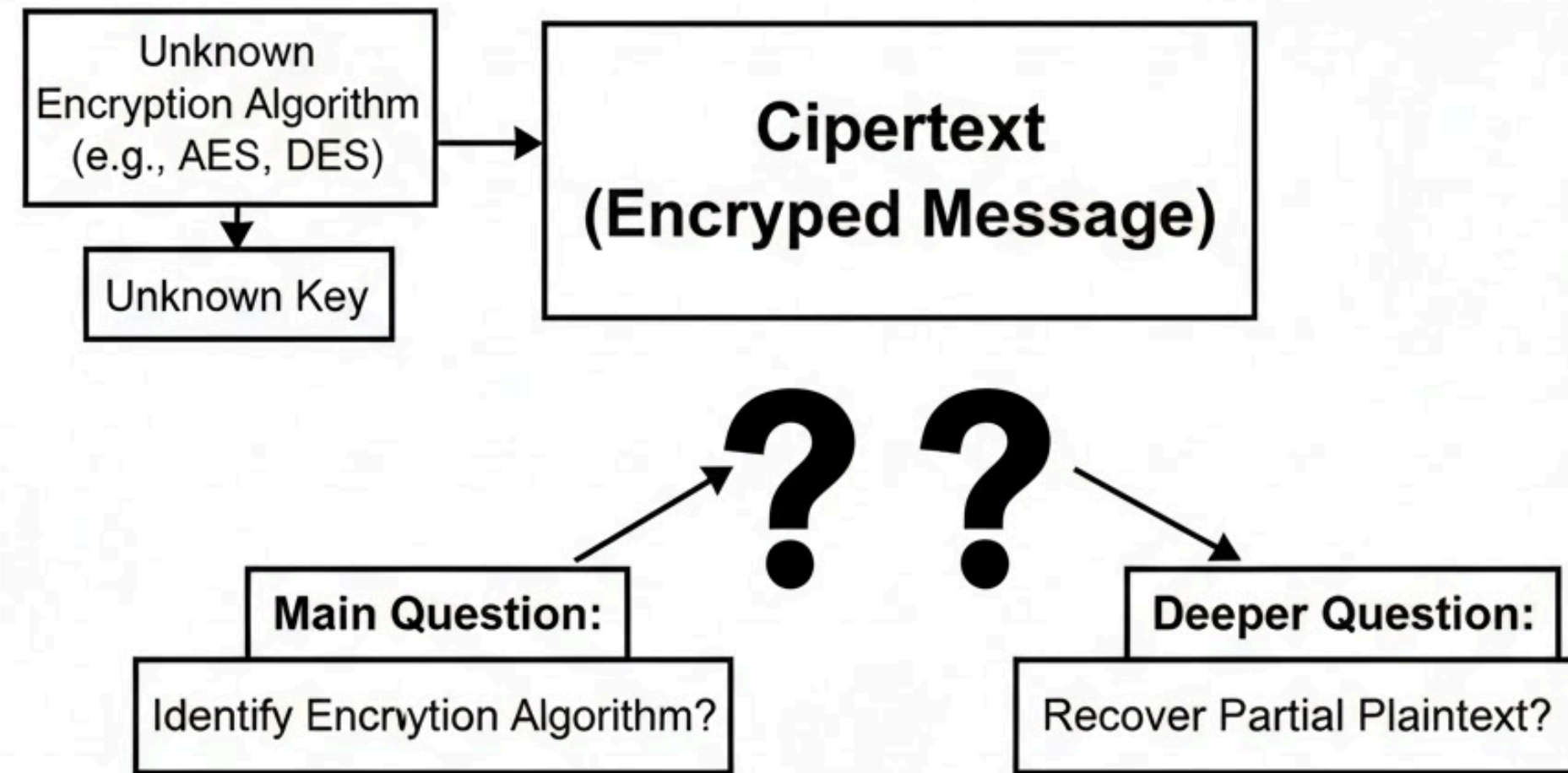


Nithin S, 221IT085

Department of Information Technology
National Institute of Technology Karnataka, Surathkal

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The Problem We're Solving



Can we use Machine Learning to automatically identify the encryption algorithm just by looking at the final encrypted message?

If we can identify the algorithm, can we also recover any part of the original plaintext message?

Why should we solve it ?

- **Digital Forensics:** When investigators find an encrypted device, their first step is to figure out what kind of encryption they're up against. This project automates that first step.
- **Security Auditing:** We can test how "leaky" an encryption algorithm is. Does it leave behind a unique "fingerprint" that our models can find? This helps us measure the actual security of a cipher.
- **Network Security:** This can help in analyzing network traffic to identify the types of encryption being used, which can be useful for threat detection.
- **Speed & Scale:** Using ML is much faster and more scalable than relying on a few human experts to manually inspect the data.

Dataset

Created a In House Dataset

- **Total Size:** A dataset of 3.5 Million encrypted files.
- **Balanced Classes:** We generated exactly 500,000 samples for each of the following 7 algorithms in TLS Cipher Suite:
 - AES
 - 3DES
 - DES
 - IDEA
 - Camellia
 - RC4
 - ChaCha20

Table 4.1: Dataset Structure Overview

ID	Algorithm	Key	IV/Nonce	Ciphertext
000001	AES-128 (CBC)	128-bit Rand	128-bit Rand IV	b9 3a f1 ... 7c
000002	RC4	128-bit Rand	N/A	ac 10 d5 ... e1
000003	3DES (CBC)	168-bit Rand	64-bit Rand IV	4f 22 91 ... 99
000004	DES (CFB)	56-bit Rand	64-bit Rand IV	a1 8c 04 ... 5b
000005	Camellia-128 (CTR)	128-bit Rand	128-bit nonce	67 2e 13 ... d4
000006	IDEA (CBC)	128-bit Rand	64-bit Rand IV	22 4f a0 ... 8c
000007	ChaCha20	256-bit Rand	96-bit nonce	ca 71 44 ... b3

Dataset

Every single one of the 3.5 million files is a completely new encryption operation.

Here is the process for making just two files in the dataset:

To create file_1 (AES):

1. Generate plaintext_A (1KB of random data).
2. Generate key_A (a new random key).
3. Encrypt to get ciphertext_A.
4. Save ciphertext_A with the label "AES".

To create file_2 (RC4):

1. Generate plaintext_B (a different 1KB of random data).
2. Generate key_B (a different new random key).
3. Encrypt to get ciphertext_B.
4. Save ciphertext_B with the label "RC4".

We did this 3.5 million times to get our 500,000 samples for each of the 7 algorithms.

Every plain text was encrypted with a new, random key and IV/nonce. The model cannot "cheat" by learning a key.

Dataset

For block ciphers (like AES, DES, 3DES, Camellia, and IDEA), we used secure, common modes:

- CBC (Cipher Block Chaining)
- CFB (Cipher Feedback)
- CTR (Counter)

We explicitly excluded ECB (Electronic Codebook) mode because it's known to be insecure and "leaky," which would make the problem too easy.

In this project, RC4 and ChaCha20 were the two stream ciphers we included.

Approach 1 : Ciphertext-Only TLS Cipher Suite Identification Using Standard Machine Learning Models

Feature Extraction: 40 NIST Randomness numeric features taken from ciphertext

Goal: To Capture patterns in ciphertext that differ by algorithm.

Table 4.2: NIST Statistical Test Suite Overview

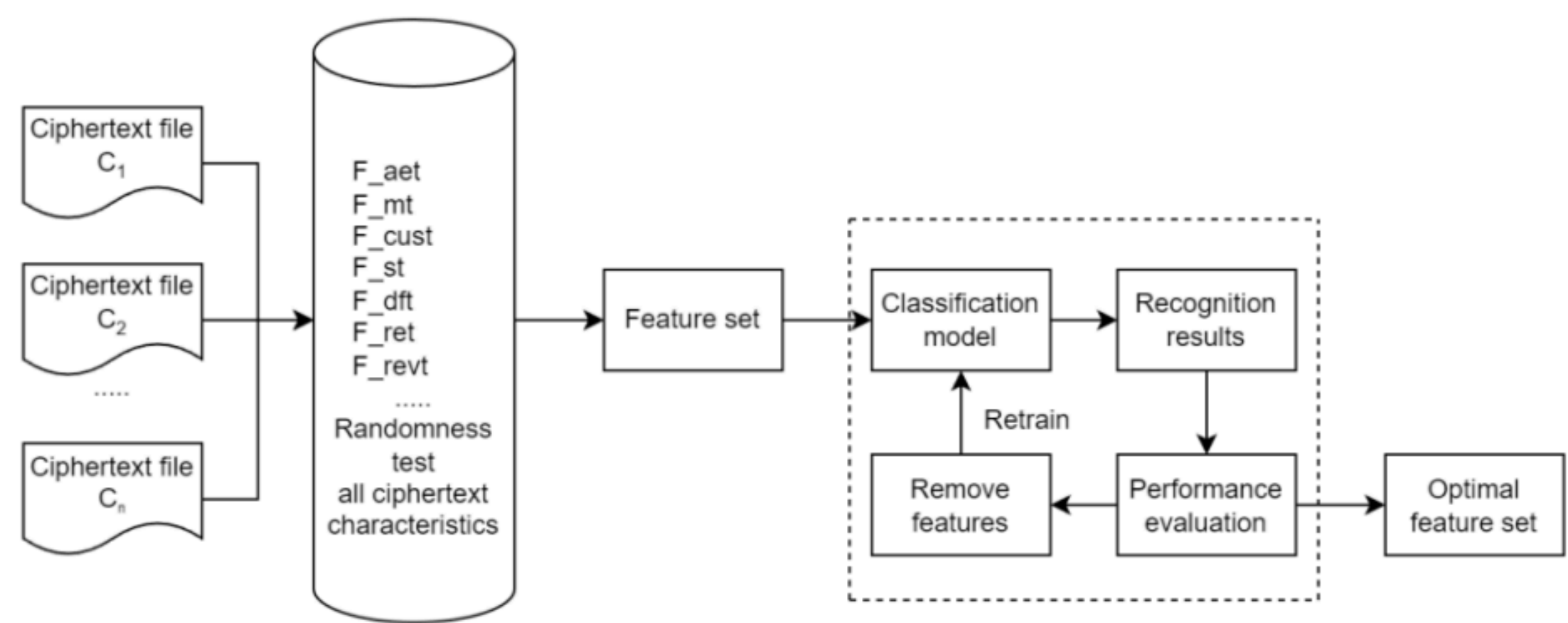
Test Name	Description
Frequency (Monobit)	Measures proportion of ones vs zeros in bit sequence
Block Frequency	Tests uniformity of ones proportion across fixed-size blocks
Runs	Detects oscillation rate between consecutive identical bits
Longest Run of Ones	Examines maximum consecutive ones length distribution
Binary Matrix Rank	Checks for linear dependence in binary matrix representations
Discrete Fourier Transform (DFT)	Identifies periodic patterns via frequency domain analysis
Non-overlapping Template	Searches for specific non-overlapping bit patterns
Overlapping Template	Detects specific overlapping bit sequences
Maurer’s Universal	Measures sequence compressibility and pattern repetition
Linear Complexity	Computes shortest LFSR that generates the sequence
Serial	Tests frequency distribution of overlapping m -bit patterns
Approximate Entropy	Quantifies regularity and predictability of patterns
Cumulative Sums	Detects bias toward sequence beginning or end
Random Excursions	Analyzes random walk state visit frequencies
Random Excursions Variant	Counts total visits to random walk states

Example

Cipher Text: "XYZ" -----> 010110000101100101011010

Feature	Meaning	Value
Bit Count (1s)	Count of 1 bits	11
Bit Count (0s)	Count of 0 bits	13
Bit Transitions	Number of flips 0→1 or 1→0	16 flips

Feature Selection: Random Forest used to rank features
Kept top 10 features



Rank	NIST Test Feature
1	Frequency (Monobit) Test
2	Block Frequency Test
3	Runs Test
4	Longest Run of Ones Test
5	Discrete Fourier Transform (DFT) Test
6	Linear Complexity Test
7	Serial Test (first p-value)
8	Serial Test (second p-value)
9	Approximate Entropy Test
10	Maurer's Universal Statistical Test

Figure 4.1: Feature Selection Workflow

Seven ML models tested using the top 10 features:

- Logistic Regression
- Decision Tree
- k-Nearest Neighbors
- SVM
- Random Forest
- AdaBoost
- Hybrid RF + Logistic Regression (best)

Train: 80%

Validation: 10%

Test: 10%

Stratified split

Results:

Binary Classification

Cipher Pair	Model	Accuracy (%)	Precision (%)	Recall (%)	F1 (%)
RC4 vs ChaCha20	HRFLR	75.8	75.6	76.1	75.8
	RF	73.2	72.8	73.5	73.1
	AdaBoost	71.9	71.6	72.3	71.9
	SVM	69.7	69.4	70.1	69.7
	LR	65.3	65.0	65.7	65.3
	KNN	62.8	62.4	63.1	62.7
	DT	60.4	60.1	60.8	60.4
AES vs 3DES	HRFLR	75.2	74.9	75.6	75.2
	RF	72.8	72.4	73.2	72.8
	AdaBoost	70.6	70.3	71.0	70.6
	SVM	68.9	68.5	69.2	68.8
	LR	64.7	64.3	65.0	64.6
	KNN	61.5	61.2	61.9	61.5
	DT	59.8	59.4	60.1	59.7
DES vs IDEA	HRFLR	64.2	64.0	64.5	64.2
	RF	61.7	61.4	61.9	61.6
	AdaBoost	60.3	60.1	60.6	60.3
	SVM	58.4	58.2	58.7	58.4
	LR	55.2	54.9	55.5	55.2
	KNN	53.1	52.8	53.4	53.1
	DT	51.9	51.6	52.2	51.9

Results:

Binary Classification

Model	Accuracy (%)	Precision (%)	Recall (%)	F1 (%)
HRFLR	70.9	70.7	71.2	70.9
RF	68.4	68.2	68.5	68.3
AdaBoost	66.8	66.5	67.1	66.8
SVM	65.1	64.8	65.4	65.1
LR	61.5	61.2	61.8	61.5
KNN	58.9	58.6	59.2	58.9
DT	57.3	57.0	57.6	57.3

Results:

3 Class Classification

Algorithm Triplet	Model	Acc (%)	Prec (%)	Rec (%)	F1 (%)
RC4, AES, ChaCha20	HRFLR	54.3	54.7	54.0	54.3
	RF	52.1	52.4	51.8	52.0
	AdaBoost	51.0	51.3	50.7	51.0
	SVM	49.8	50.1	49.5	49.8
	LR	46.2	46.5	45.9	46.2
	KNN	44.1	44.3	43.8	44.0
	DT	42.7	42.9	42.5	42.7
DES, 3DES, IDEA	HRFLR	49.8	50.1	49.5	49.8
	RF	47.7	47.9	47.4	47.6
	AdaBoost	46.5	46.8	46.3	46.5
	SVM	45.1	45.4	44.9	45.1
	LR	41.9	42.1	41.6	41.8
	KNN	39.7	39.9	39.5	39.7
	DT	38.3	38.5	38.1	38.3

Results:

3 Class Classification

Model	Accuracy (%)	Precision (%)	Recall (%)	F1 (%)
HRFLR	52.4	52.7	52.1	52.4
RF	50.3	50.4	50.1	50.2
AdaBoost	49.1	49.3	48.9	49.1
SVM	47.6	47.8	47.4	47.6
LR	44.2	44.4	44.0	44.2
KNN	42.1	42.3	41.9	42.1
DT	40.8	40.9	40.6	40.7

Results:

6 Class Classification

Model	Accuracy (%)	Precision (%)	Recall (%)	F1 (%)
HRFLR	33.2	33.5	32.8	33.1
RF	31.4	31.1	30.9	31.0
AdaBoost	30.1	30.3	29.8	30.0
SVM	28.7	28.9	28.4	28.6
LR	26.8	26.5	26.2	26.3
KNN	24.9	24.6	24.3	24.4
DT	23.1	22.8	22.6	22.7
<i>Random Baseline</i>	<i>16.7</i>	<i>16.7</i>	<i>16.7</i>	<i>16.7</i>

Observation

Scenario	Random	HRFLR	RF	AdaBoost	SVM	LR	KNN	DT
Binary (mean)	50.0%	70.9%	68.4%	66.8%	65.1%	61.5%	58.9%	57.3%
Three-Class	33.3%	52.4%	50.3%	49.1%	47.6%	44.2%	42.1%	40.8%
Six-Class	16.7%	33.2%	31.4%	30.1%	28.7%	26.8%	24.9%	23.1%

- Ciphertext-only identification is possible, but accuracy is limited
- Best performance seen in binary cases (~71%)
- Accuracy drops sharply for 3-class (~52%) and 6-class (~33%) tasks
- Ensemble models (Hybrid RF-LR, Random Forest) clearly outperform others
- Stream ciphers (ChaCha20, RC4) show more distinct patterns than block ciphers
- Legacy block ciphers (DES, 3DES, IDEA) are hardest to distinguish
- Top useful features: entropy, randomness tests, bit-transition patterns
- Results confirm: modern cryptography hides patterns effectively

Approach 2 : Ciphertext-Only TLS Cipher Suite Identification Using Multi Layered Perceptron

Feature Extraction: 40 NIST Randomness numeric features taken from ciphertext

Goal: To Capture patterns in ciphertext that differ by algorithm.

Table 4.2: NIST Statistical Test Suite Overview

Test Name	Description
Frequency (Monobit)	Measures proportion of ones vs zeros in bit sequence
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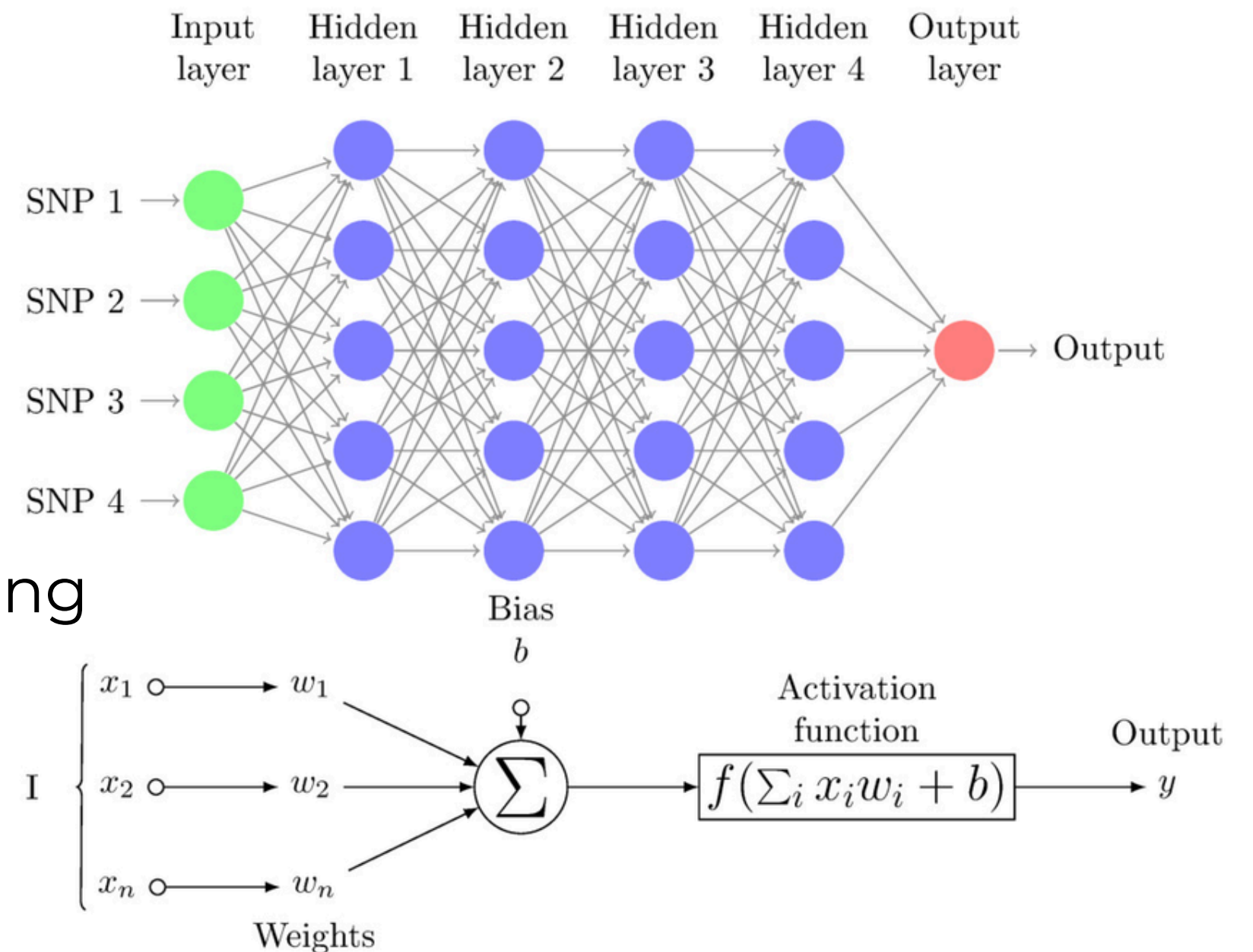
Dimensionality Reduction

- PCA applied to shrink feature vector
- Reduced from 40 → 10 features
- ~98% variance retained
- Helps speed up learning and reduce noise

MLP Model Design

- RELU activation
- Batch normalization
- Dropout (0.2) to avoid overfitting
- AdamW optimizer with learning rate scheduling
- Trained for 15–25 epoch

- Dataset: 3.5M samples
- Train / Val / Test split: 80% / 10% / 10%
- Large batch size: 4096
- Early stopping based on F1-score



Results:

Binary Classification

Cipher Pair	Accuracy (%)	Macro-F1	ROC-AUC	Rank
3DES vs DES	73.4	0.728	0.799	1
AES vs DES	73.1	0.724	0.796	2
ChaCha20 vs DES	72.9	0.723	0.792	3
AES vs RC4	72.6	0.720	0.792	4
RC4 vs DES	71.7	0.710	0.785	5
Camellia vs DES	71.6	0.709	0.782	6
AES vs ChaCha20	71.2	0.707	0.781	7
ChaCha20 vs 3DES	70.8	0.700	0.777	8
DES vs IDEA	70.6	0.699	0.776	9
Camellia vs RC4	70.5	0.698	0.775	10
AES vs 3DES	70.1	0.696	0.768	11
RC4 vs 3DES	69.5	0.686	0.766	12
Camellia vs ChaCha20	69.2	0.684	0.764	13
AES vs IDEA	68.9	0.681	0.757	14
3DES vs IDEA	68.2	0.673	0.755	15
ChaCha20 vs IDEA	67.8	0.670	0.753	16
Camellia vs 3DES	67.4	0.667	0.748	17
RC4 vs IDEA	66.9	0.661	0.748	18
Camellia vs IDEA	66.1	0.653	0.742	19
AES vs Camellia	63.8	0.629	0.734	20
ChaCha20 vs RC4	62.7	0.618	0.721	21
Mean	69.1	0.683	0.765	—

Results:

Binary Classification

Model	Accuracy (%)	Macro-F1	ROC-AUC	from MLP
MLP (Primary)	69.1	0.683	0.765	—
SVM (RBF)	64.2	0.634	0.718	-4.9
Random Forest	63.1	0.623	0.706	-6.0
Logistic Regression	60.7	0.599	0.681	-8.4
k -NN ($k = 15$)	58.9	0.581	0.663	-10.2
Gaussian Naive Bayes	55.4	0.546	0.627	-13.7

Results:

3 Class Classification

Algorithm Triplet	Acc (%)	Macro-F1	Macro-AUC	Rank
AES, 3DES, DES	52.3	0.517	0.765	1
AES, ChaCha20, RC4	51.8	0.510	0.756	2
AES, RC4, DES	51.1	0.504	0.752	3
Camellia, 3DES, DES	50.7	0.499	0.748	4
AES, IDEA, 3DES	50.6	0.498	0.746	5
ChaCha20, RC4, 3DES	50.2	0.496	0.744	6
AES, Camellia, ChaCha20	49.6	0.489	0.742	7
ChaCha20, RC4, DES	49.1	0.483	0.739	8
Camellia, IDEA, DES	48.8	0.479	0.737	9
AES, Camellia, IDEA	47.9	0.470	0.731	10

Results:

3 Class Classification

Table 1: Performance of various models on 3-class classification task				
Model	Accuracy (%)	Macro-F1	Macro-AUC	from MLP
MLP (Primary)	50.4	0.496	0.742	—
SVM (RBF)	45.0	0.442	0.694	-5.4
Random Forest	44.3	0.435	0.687	-6.1
Logistic Regression	41.6	0.408	0.661	-8.8
k -NN ($k = 15$)	40.8	0.400	0.653	-9.6
Gaussian Naive Bayes	37.2	0.364	0.617	-13.2
Random Baseline	33.3	0.333	0.500	-17.1

Results:

6 Class Classification

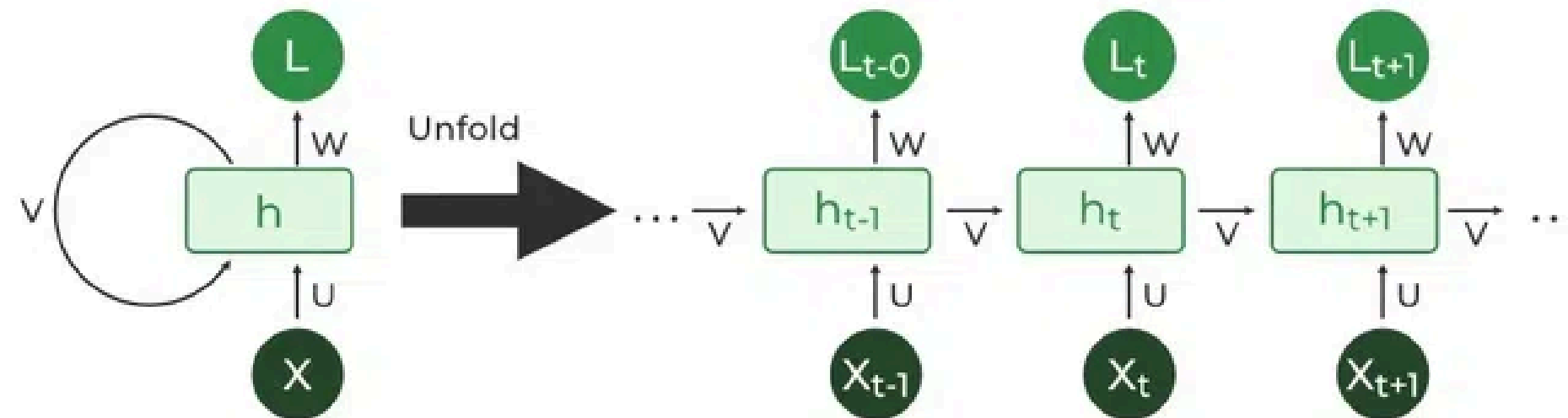
Model	Accuracy (%)	Macro-F1	Macro-AUC	from MLP
MLP (Primary)	30.8	0.301	0.706	—
SVM (RBF)	26.4	0.258	0.661	-4.4
Random Forest	25.7	0.251	0.654	-5.1
Logistic Regression	23.1	0.225	0.628	-7.7
k -NN ($k = 15$)	22.6	0.220	0.621	-8.2
Gaussian Naive Bayes	20.3	0.198	0.595	-10.5
<i>Random Baseline</i>	<i>16.7</i>	<i>0.167</i>	<i>0.500</i>	<i>-14.1</i>

Observation

Classification Task	MLP (%)	HRFLR (%)	Difference
Binary (mean)	69.1	70.9	-1.8
Three-Class (median)	50.4	52.4	-2.0
Six-Class	30.8	33.2	-2.4

- MLP achieves 69% / 50% / 31% for binary / 3-class / 6-class
- Outperforms classical ML in most cases
- Slightly below best hybrid ensemble model
- Confirms: ciphertext-only cipher detection remains challenging

Approach 3 : Ciphertext-Only TLS Cipher Suite Identification Using Recurrent Neural Networks



- Extends ciphertext-only detection to sequence models
- RNNs learn patterns across byte positions
- Treat ciphertext as a sequence, not just features
- Enables learning subtle temporal signal patterns
- **Goal:** identify algorithm from ciphertext stream alone

Models Evaluated

- LSTM (2-layer)
- BiLSTM (bidirectional LSTM)
- GRU (2-layer)
- 1D-CNN (temporal conv baseline)
- Random Forest (non-sequential baseline)

Model Design

- Optimizer: Adam
- LR decay + gradient clipping
- Dropout: 0.3
- Label smoothing
- Batch size: 256
- Max epochs: 50 (early stopping)
- Random byte shuffle augmentation (5%)

Results:

Binary Classification

Model	Accuracy (%)	Precision (%)	Recall (%)	F1 (%)
BiLSTM	73.4	73.1	73.0	73.0
GRU	72.1	71.9	71.8	71.7
LSTM	71.2	71.0	70.8	70.7
1D-CNN	67.8	67.2	67.0	67.1
RF	64.5	64.0	63.7	63.8
<i>Random Baseline</i>	<i>50.0</i>	<i>50.0</i>	<i>50.0</i>	<i>50.0</i>

Results:

3 Class Classification

Model	Accuracy (%)	Precision (%)	Recall (%)	F1 (%)
BiLSTM	51.4	50.8	50.5	50.7
GRU	50.2	49.7	49.5	49.6
LSTM	49.6	49.1	48.8	48.9
1D-CNN	46.8	46.1	46.0	46.0
RF	43.9	43.2	42.9	43.0
<i>Random Baseline</i>	<i>33.3</i>	<i>33.3</i>	<i>33.3</i>	<i>33.3</i>

Results:

6 Class Classification

Model	Accuracy (%)	Precision (%)	Recall (%)	F1 (%)
BiLSTM	31.1	30.8	30.6	30.6
GRU	30.2	30.0	29.8	29.8
LSTM	29.4	29.2	29.0	28.9
1D-CNN	26.9	26.3	26.0	26.2
RF	24.7	24.4	24.2	24.3
<i>Random Baseline</i>	<i>16.7</i>	<i>16.7</i>	<i>16.7</i>	<i>16.7</i>

Observation

Model	Accuracy (%)	Learning Paradigm
HRFLR (Section 5.1.11)	33.2	Tree-based meta-ensemble
BiLSTM	31.1	Bidirectional recurrent
MLP (Section 5.2.6)	30.8	Feed-forward deep learning
GRU	30.2	Gated recurrent
LSTM	29.4	Unidirectional recurrent
SVM (Section 5.1.11)	28.7	Kernel-based
1D-CNN	26.9	Convolutional
RF (Section 5.1.11)	25.7	Tree-based ensemble
<i>Random Baseline</i>	<i>16.7</i>	—

Observation

- Recurrent models extract useful sequential patterns from ciphertext
- BiLSTM achieves highest accuracy, followed closely by GRU
- Sequence learning clearly outperforms statistical and feed-forward models
- Binary case strong (~73%), but multi-class accuracy drops sharply
- GRU offers best accuracy vs computational efficiency trade-off
- 1D-CNN detects local patterns but lacks long-range context
- Overall: RNNs improve detection but ciphertext-only limits remain, especially for modern algorithms

Round-Based Leakage Evaluation

- Goal: measure cipher distinguishability across encryption rounds
- **Round checkpoints:**
 - R1: early-round leakage
 - R4: mid-round
 - R8: late-round, near full diffusion
 - LR: final output (ciphertext-only)
- Models compared: HRLFR, MLP, BiLSTM

Results:

Binary Classification with HRLFR

Cipher Pair	R1	R4	R8	LR
AES vs DES	94.6	82.1	72.4	66.3
AES vs IDEA	95.1	84.3	74.2	67.5
AES vs Camellia	93.8	81.5	71.1	65.7
AES vs RC4	96.4	89.2	78.0	69.8
AES vs ChaCha20	95.9	87.6	76.1	68.2
DES vs IDEA	94.3	82.9	72.7	66.1
DES vs Camellia	93.5	81.1	70.6	65.4
DES vs RC4	96.1	88.5	77.4	69.1
DES vs ChaCha20	95.7	87.0	75.5	68.0
IDEA vs Camellia	94.7	83.6	73.0	66.6
IDEA vs RC4	96.3	88.9	78.1	69.6
IDEA vs ChaCha20	95.8	87.4	75.8	68.1
Camellia vs RC4	96.5	89.0	77.9	69.4
Camellia vs ChaCha20	95.6	87.1	75.4	67.8
RC4 vs ChaCha20	96.8	90.2	79.0	70.3

Results:

Binary Classification with MLP

Cipher Pair	R1	R4	R8	LR
AES vs DES	97.1	86.7	77.0	70.1
AES vs IDEA	97.5	88.9	78.9	71.4
AES vs Camellia	96.8	85.9	75.6	69.3
AES vs RC4	98.2	92.1	83.5	73.8
AES vs ChaCha20	97.9	91.0	81.7	72.1
DES vs IDEA	97.0	87.2	76.5	69.8
DES vs Camellia	96.3	85.4	75.2	69.0
DES vs RC4	98.0	91.4	82.8	73.2
DES vs ChaCha20	97.7	90.2	81.0	71.4
IDEA vs Camellia	97.2	88.0	77.4	70.2
IDEA vs RC4	98.3	92.0	84.0	74.0
IDEA vs ChaCha20	97.8	90.9	81.6	72.5
Camellia vs RC4	98.4	91.8	83.0	73.3
Camellia vs ChaCha20	97.6	90.0	80.4	71.6
RC4 vs ChaCha20	98.6	93.5	84.8	74.3

Results:

Binary Classification with BiLSTM

Cipher Pair	R1	R4	R8	LR
AES vs DES	98.0	88.8	79.4	72.8
AES vs IDEA	98.3	90.6	81.2	73.9
AES vs Camellia	97.5	87.9	78.0	72.0
AES vs RC4	98.9	94.3	87.0	75.2
AES vs ChaCha20	98.6	92.8	84.4	73.6
DES vs IDEA	97.8	89.5	79.0	72.3
DES vs Camellia	97.1	87.3	77.1	71.6
DES vs RC4	98.7	93.8	86.1	74.6
DES vs ChaCha20	98.4	92.2	83.5	73.0
IDEA vs Camellia	97.9	90.1	79.8	72.4
IDEA vs RC4	98.8	94.0	86.9	75.1
IDEA vs ChaCha20	98.5	92.7	84.2	73.4
Camellia vs RC4	98.8	93.6	86.0	74.4
Camellia vs ChaCha20	98.3	92.1	83.2	73.3
RC4 vs ChaCha20	99.0	95.4	88.7	76.0

Results:

3 Class Classification

Model	R1	R4	R8	LR
HRLFR	88.6	78.0	66.3	52.8
MLP	92.4	82.9	72.1	56.4
BiLSTM	94.1	85.7	75.3	59.8

Results:

6 Class Classification

Model	R1	R4	R8	LR
HRLFR	64.8	48.3	38.2	30.1
MLP	68.9	53.7	42.8	32.4
BiLSTM	71.2	56.9	46.6	33.8

Observation

- Early-round crypto outputs are highly distinguishable
- As encryption rounds increase $\rightarrow$ ML accuracy drops sharply
- Final-round identification accuracy approaches lower theoretical bound
- Confirms strong diffusion properties of secure modern ciphers
- Demonstrates that round depth is essential for hiding algorithm fingerprints

Plaintext Recovery

- Go beyond algorithm identification
- Test if ciphertext-only setting leaks enough signal to reconstruct plaintext
- Evaluate **neural seq-to-seq** models on multiple ciphers
- Explore leakage across legacy and modern designs

Model Setup

- Encoder–decoder LSTM with attention
- 3-layer encoder (512 units)
- 3-layer decoder (512 units)

Model Setup

- Input: ciphertext sequences
- Output: predicted plaintext tokens
- Ciphers tested:
 - DES (legacy)
 - Twofish (moderate legacy-modern transition)
 - AES (modern block cipher)
 - ChaCha20 (modern stream cipher)

Evaluation

Cipher	Model	F1	Levenshtein	ROUGE-1	ROUGE-2	ROUGE-L
DES	Seq2Seq LSTM	0.872	0.0408	0.790	0.694	0.790
Twofish	Seq2Seq LSTM	0.751	0.0631	0.710	0.633	0.710
AES	Seq2Seq BiLSTM	0.312	0.2650	0.220	0.078	0.220
ChaCha20	Seq2Seq LSTM	0.208	0.3240	0.150	0.041	0.140

Evaluation

- Seq-to-seq RNNs with attention reveal major leakage in legacy ciphers
- Twofish shows noticeable leakage, though less severe than DES
- AES and ChaCha20 remain secure, with recovered output resembling noise
- Recovery quality strongly correlates with diffusion strength and modern cipher design
- Learned models exploit round structure, padding behavior, and byte-frequency bias when available
- Demonstrates neural cryptanalysis can expose weak primitives, but
- cannot break contemporary encryption purely from ciphertext

THANK YOU